

Hailey Garcia

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Department of Mathematics, Louisiana State University, Baton Rouge, LA 70803

Education

Ph.D. Mathematics, Louisiana State University	Fall 2025–present
<i>Advisor: Christin Bibby</i>	
B.S. Mathematics, Louisiana State University	Fall 2022–Spring 2025

Papers and Preprints

G. Agrahari, C. Bibby, S. Boros, H. J. Garcia, F. Heidercheidt, Z. Wang. *On some structural properties of graphs with non-negative resistance curvature*. (2026). [arXiv:2607.13169](https://arxiv.org/abs/2607.13169).

M. Fairchild, H. J. Garcia, J. Murphy, H. Percle. *On the non-orientable 4-genus of torus knots with braid index 5 or 6*. *Journal of Knot Theory and Its Ramifications* **35** (2026), No. 7, 2650001. <https://doi.org/10.1142/S021821652650001X>.

Talks

Noncontractive Function Systems	Spring 2023
<i>6th TX-LA Mathematics Undergraduate Conference</i>	
On The “Kelly Sequence”	Spring 2023
<i>6th TX-LA Mathematics Undergraduate Conference</i>	

Teaching

MATH 1431, Business Calculus — Lab Leader/TA, LSU	Spring 2026
<i>Sections 7, 11, 19</i>	
MATH 1431, Business Calculus — Lab Leader/TA, LSU	Fall 2025
<i>Sections 7, 8, 9, 11, 25, 28</i>	
Honors Calculus 1 — Primary Instructor, Arden Cahill Academy	Fall 2023–Spring 2024

Mentoring

Mentor, Directed Reading Program	Fall 2026
TA, BSM REU at Haverford College	Summer 2026
TA, 2026 LSU Undergraduate FRC	Summer 2026
Mentor, Vertically Integrated Research Courses, LSU	Fall 2025–present

Service

Webmaster, LSU AMS Student Chapter	Fall 2026–present
Webmaster, LSU RTG	Spring 2026–present
Manager, LSU Math 3D Print Lab	Spring 2026–present
Officer, LSU AWM Student Chapter	Fall 2025–present
<i>President (Fall 2025–Spring 2026); Secretary (Spring 2026–present); Webmaster (Fall 2025–present)</i>	

Fellowships and Awards

LSU RTG Fellow	Fall 2026–present
Pasquale Porcelli Graduate Student Academic Excellence Award	Spring 2026
Robert and Betti Giles Senior Mathematics Award	Spring 2025
Peg and Tom Madden Mathematics Research Scholarship	Spring 2024
Porcelli Junior Scholarship	Spring 2023